

INT 422
Pattern Recognition

2024

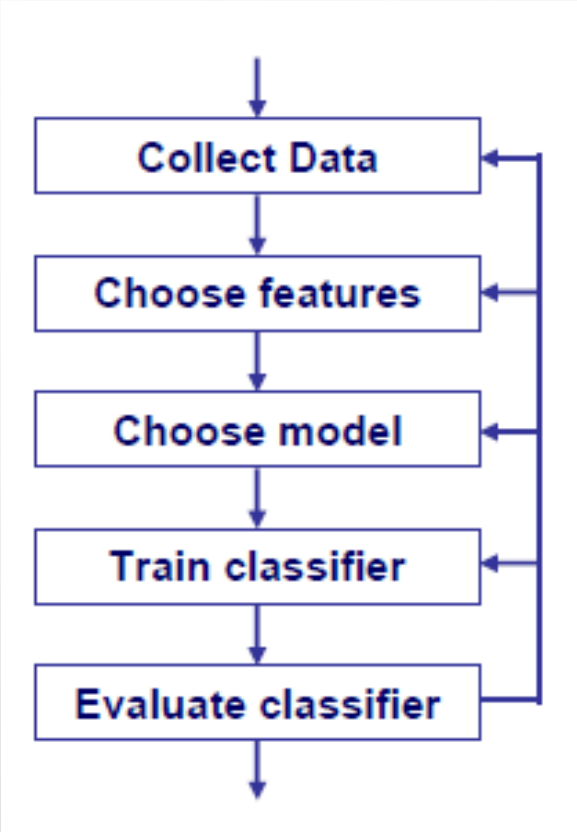
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Lecture 2-3: Agenda

- The concept of features
- Examples
- What is Bayes probability function
- Why learn the Bayes probability function
- priori probability, class-conditional probability density, posteriori probability
- Bayes formula
- Examples

The Design Cycle

- Data collection
- Feature choice
- Model choice
- Training
- Evaluation
- Computational complexity

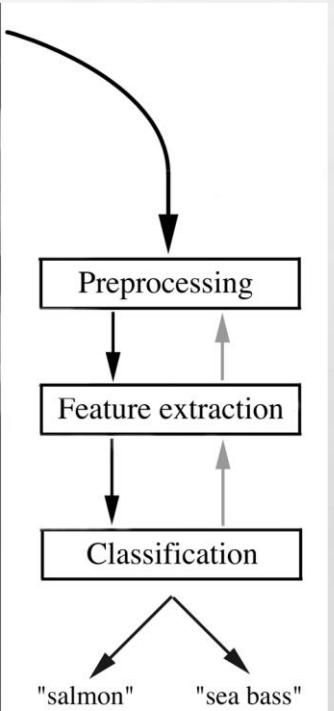


The Concept of Features

- Problem: sorting incoming fish on a conveyor belt according to species.
- Assume that we have only two kinds of fish:
  - **Sea bass.**
  - **Salmon.**

https://www.youtube.com/watch?v=xTiZ3v_ZsVo





An example: decision process

- What kind of information can distinguish one species from the other?
  - Length, width, weight, number, shape of fins, tail shape, etc.
- What can cause problems during sensing?
  - Lighting conditions, position of fish on the conveyor belt, camera noise, etc.
- What are the steps in the process?
  - **Capture image --> isolate fish --> take measurements --> make decision**

An example: Selecting Features

- Assume a fisherman told us that a sea bass is generally longer than a salmon.
- We can use length as a feature and decide between sea bass and salmon according to a threshold on length.
- How can we choose this **threshold**?

Fish length as a feature

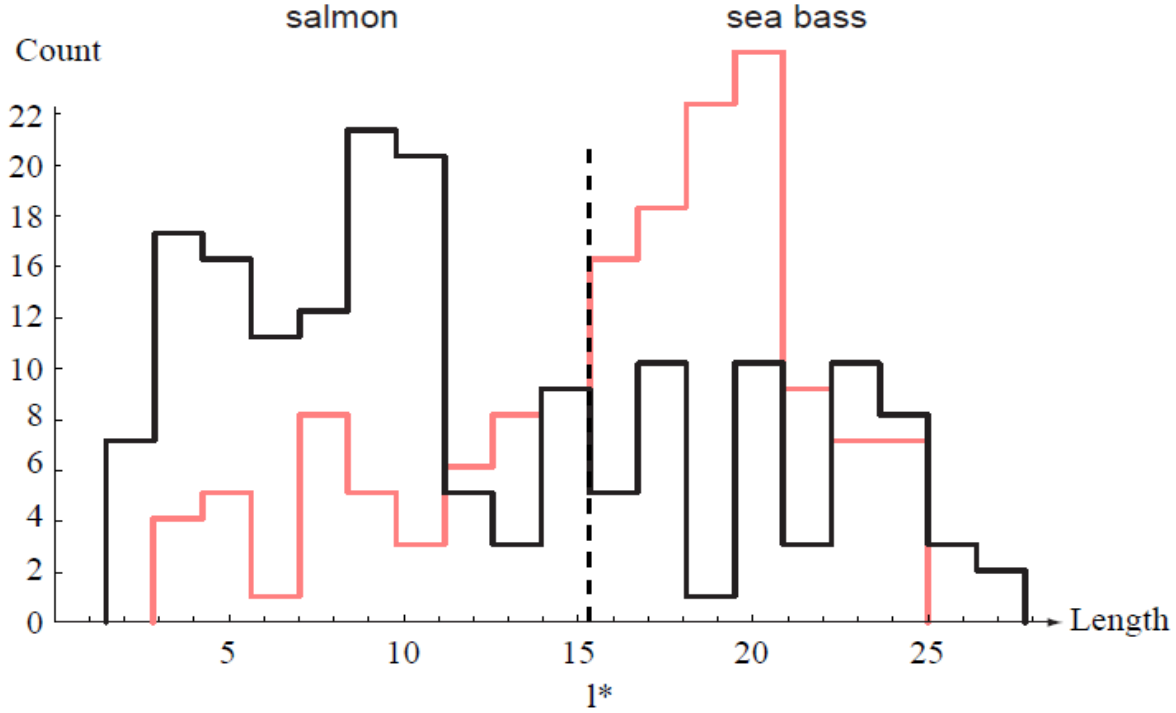


Figure 1.2: Histograms for the length feature for the two categories. No single threshold value l^* (decision boundary) will serve to unambiguously discriminate between the two categories; using length alone, we will have some errors. The value l^* marked will lead to the smallest number of errors, on average.

Duda, Richard O., Peter E. Hart, and David G. Stork. *Pattern classification*. John Wiley & Sons, copyright 2001.

Fish lightness as a feature

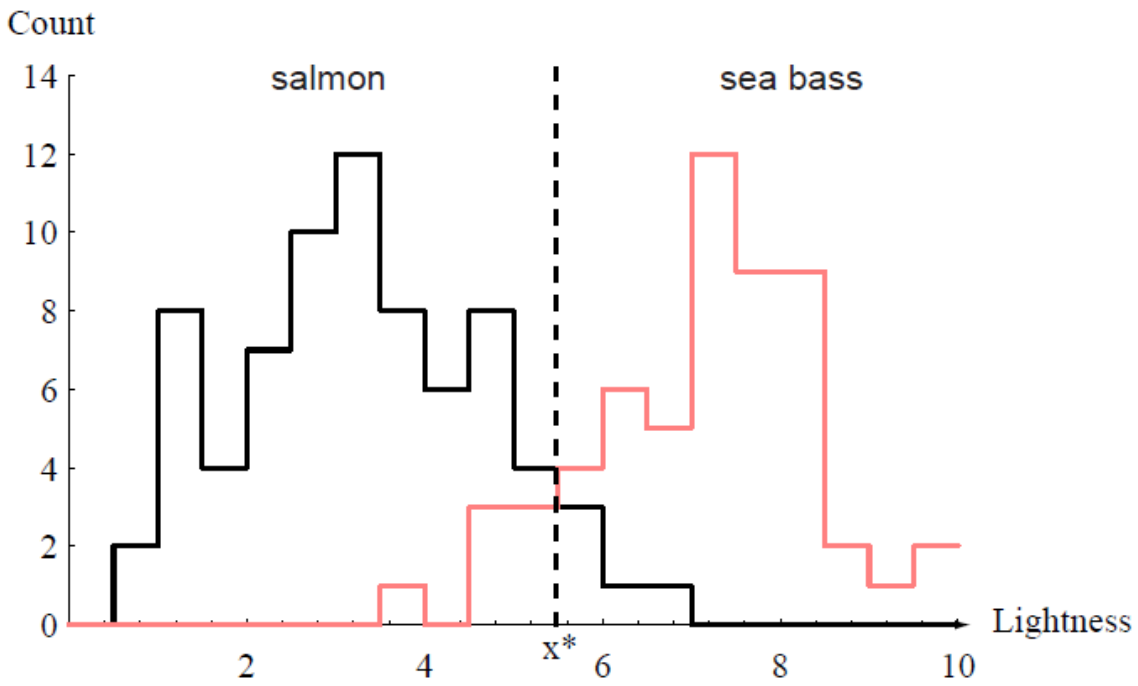


Figure 1.3: Histograms for the lightness feature for the two categories. No single threshold value x^* (decision boundary) will serve to unambiguously discriminate between the two categories; using lightness alone, we will have some errors. The value x^* marked will lead to the smallest number of errors, on average.

Duda, Richard O., Peter E. Hart, and David G. Stork. *Pattern classification*. John Wiley & Sons, copyright 2001.

Two-dimensional feature space

duda, richard o., peter e. hart, and david g. stork. *pattern classification*. john wiley & sons, copyright 2001.

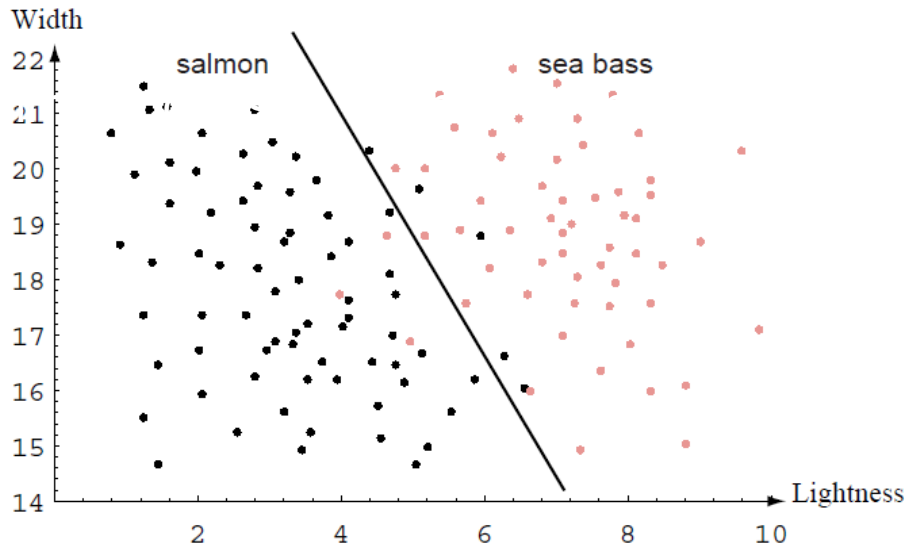


Figure 1.4: The two features of lightness and width for sea bass and salmon. The dark line might serve as a decision boundary of our classifier. Overall classification error on the data shown is lower than if we use only one feature as in Fig. 1.3, but there will still be some errors.

Joint distribution of two features leads to better separation

Complex decision boundary (polynominal classifier)

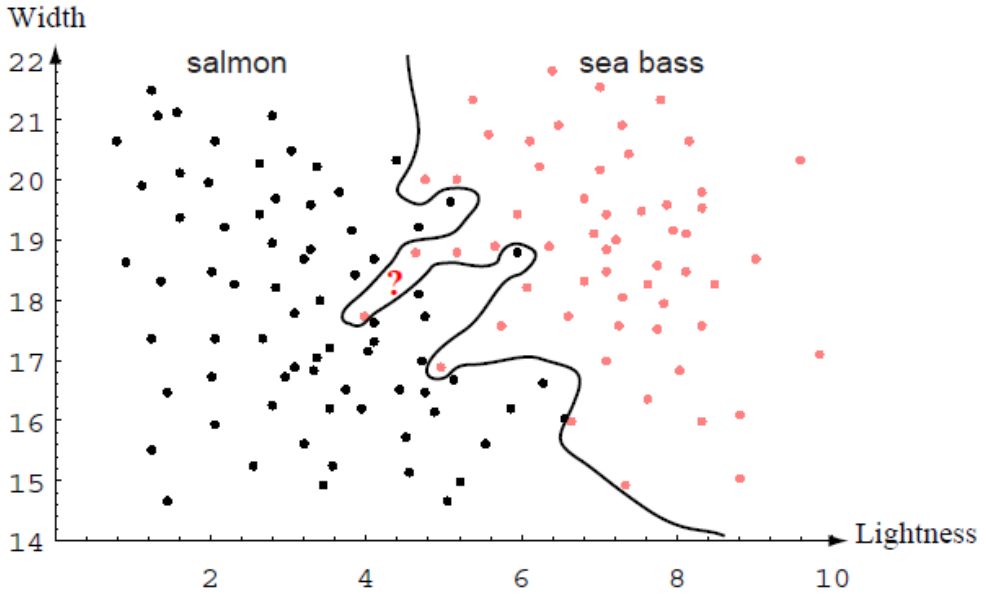


Figure 1.5: Overly complex models for the fish will lead to decision boundaries that are complicated. While such a decision may lead to perfect classification of our training samples, it would lead to poor performance on future patterns. The novel test point marked ? is evidently most likely a salmon, whereas the complex decision boundary shown leads it to be misclassified as a sea bass.

Duda, Richard O., Peter E. Hart, and David G. Stork. *Pattern classification*. John Wiley & Sons, copyright 2001.

What is the generalization ability of the classifier?

Good generalization & good accuracy

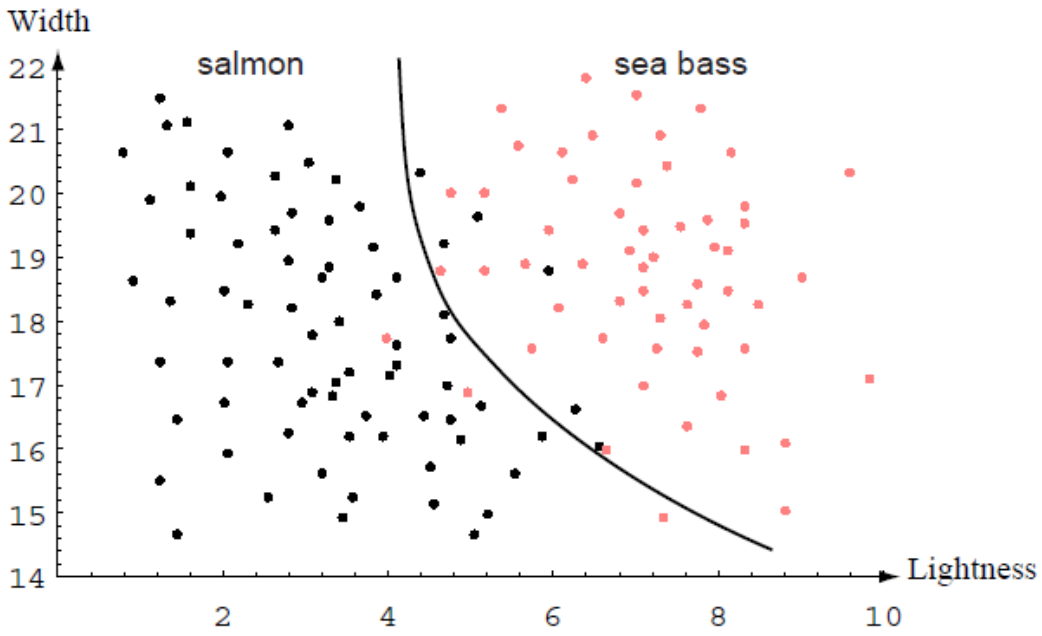


Figure 1.6: The decision boundary shown might represent the optimal tradeoff between performance on the training set and simplicity of classifier.

Duda, Richard O., Peter E. Hart, and David G. Stork. *Pattern classification*. John Wiley & Sons, copyright 2001.

Simple decision boundaries are preferred

An example: Multiple Features

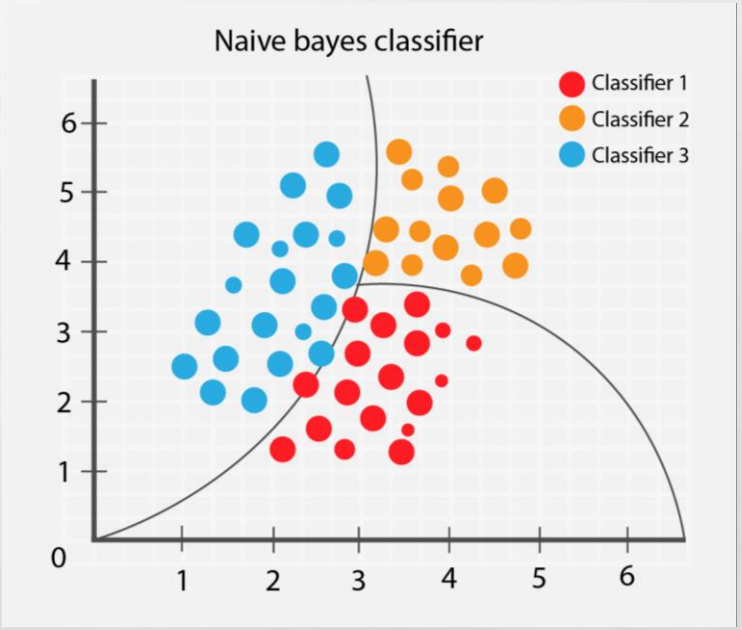
- **Does adding more features always improve the results?**
  - Avoid unreliable features.
  - Be careful about correlations with existing features.
  - Be careful about measurement costs.
  - Be careful about noise in the measurements.
- **Is there some curse for working in very high dimensions?**

The Golden Rule

"If You Have Two Equally Likely Solutions to a Problem, Choose The Simplest"

Bayesian Decision Theory

- Bayesian decision theory is a fundamental statistical approach to the problem of pattern classification.



What Is Bayes Theorem In Probability?

- In the field of probability, bayes theorem refers to a mathematical formula.
- This formula is used to calculate the conditional probability of a specific event.
- Conditional probability is nothing but the possibility of occurrence of any particular event, which is based on the outcome of an event that has already taken place.
- In calculating the conditional probability of an event, bayes theorem considers the knowledge of all conditions related to that event.
- So, if we are already aware of the conditional probability, it becomes easier to calculate the reverse probabilities with the help of bayes theorem.

Is Bayes Theorem Useful In Machine Learning?

- Bayes theorem is **extensively applied in machine learning and artificial intelligence projects.**
- It offers a way to connect a machine-learning model with an available dataset.
- Bayes theorem provides a probabilistic model that describes the association between a hypothesis and data.
- You can consider a machine learning model or algorithm as a specific framework that explains the structured associations in the data.
- So using Bayes theorem in applied machine learning, you can test and analyze different hypotheses or models based on different sets of data and calculate the probability of a hypothesis based on its prior probability.

16

What Are The Most Popular Bayesian Machine Learning Applications?

- In data analytics, Bayesian machine learning is one of the most powerful tools available to data scientists.
- **One of the most fantastic examples of real-world Bayesian machine learning applications is detecting credit card fraud.**
- Bayesian machine learning algorithms can help detect patterns that suggest potential credit card frauds.
- Bayes theorem in machine learning is also used **in advanced medical diagnosis and calculates the probability of patients developing a specific ailment based on their previous health data.**
- Other significant applications include **teaching robots to make decisions, predicting the weather, recognizing emotions from speech**, etc.

17

Bayesian Decision Theory

- Bayesian Decision Theory is a fundamental statistical approach that quantifies the tradeoffs between various decisions using probabilities and costs that accompany such decisions.
- First, we will assume that all probabilities are known.
- Then, we will study the cases where the probabilistic structure is not completely known.

Fish Sorting Example Revisited

- State of nature is a random variable.
- Define w as the type of fish we observe (state of nature, *class*) where
  - $w = w_1$ for sea bass,
  - $w = w_2$ for salmon.
  - $P(w_1)$ is the *a priori probability* that the next fish is a sea bass.
  - $P(w_2)$ is the a priori probability that the next fish is a salmon.

Prior Probabilities

- Prior probabilities reflect our knowledge of how likely each type of fish will appear before we actually see it.
- How can we choose $p(w_1)$ and $p(w_2)$?
  - Set $p(w_1) = p(w_2)$ if they are equiprobable (*uniform priors*).
  - May use different values depending on the fishing area, time of the year, etc.
- Assume there are no other types of fish

$$P(W1) + P(W2) = 1$$

(Exclusivity and exhaustivity).

Making a Decision

- How can we make a decision with only the prior information?

$$\text{Decide} \begin{cases} w_1 \text{ if } P(w_1) > P(w_2) \\ W_2 \text{ otherwise} \end{cases}$$

- What is the *probability of error* for this decision?

$$P(error) = \min\{P(w_1), P(w_2)\}$$

Class-Conditional Probabilities

- Let's try to improve the decision using the lightness measurement x.
- Let x be a continuous random variable.
- Define $p(x|w_j)$ as the *class-conditional probability density* (probability of x given that the state of nature is w_j for $j = 1, 2$).
- $p(x|w_1)$ and $p(x|w_2)$ describe the difference in lightness between populations of sea bass and salmon.

Class-conditional Probabilities

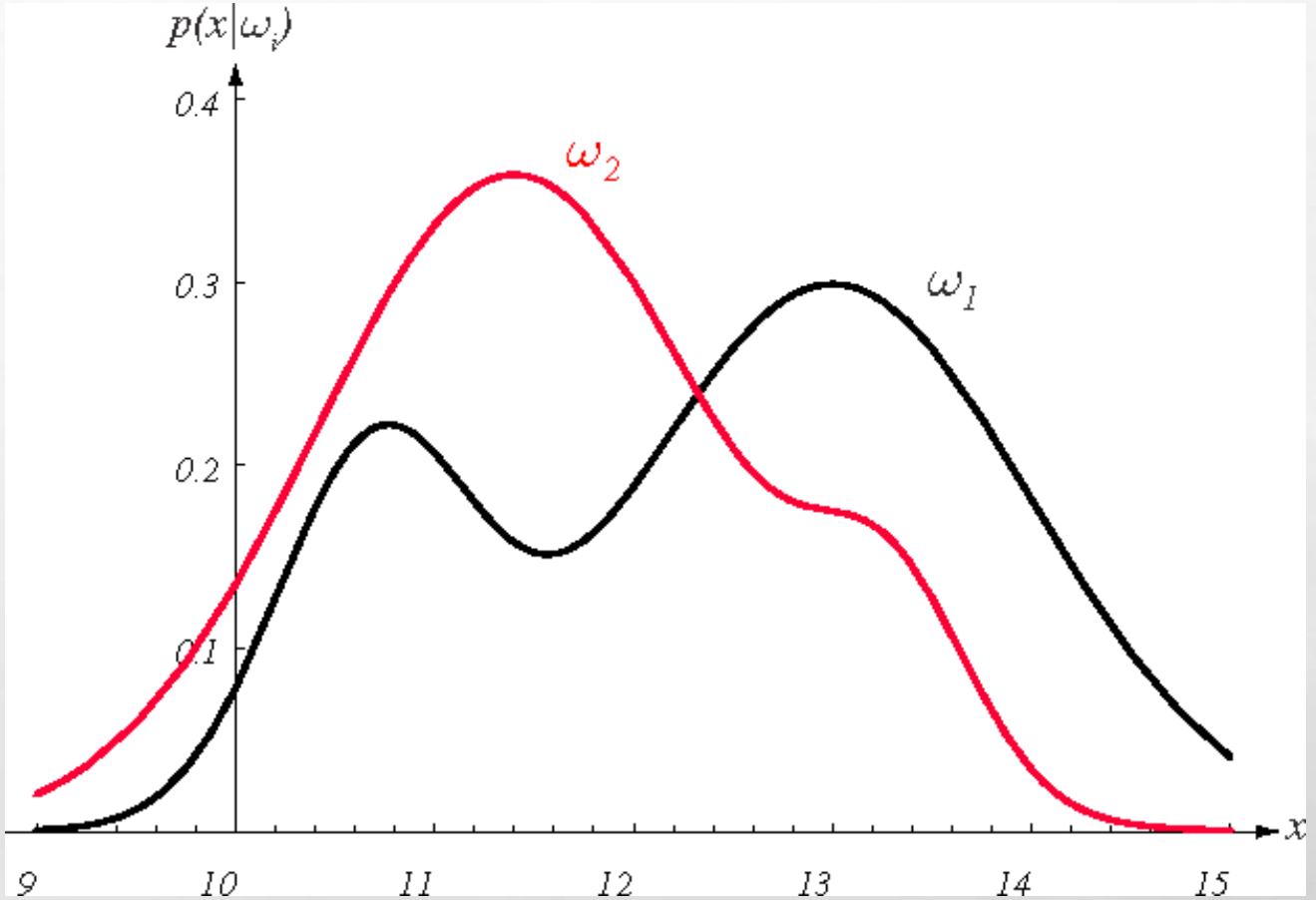


Figure 1: Hypothetical class-conditional probability density functions for two classes.

Posterior Probabilities

- Suppose we know $P(w_j)$ and $p(x|w_j)$ for $j = 1, 2$, and measure the lightness of a fish as the value x.
- Define $P(w_j|x)$ as the *a posteriori probability* (probability of the state of nature being w_j given the measurement of feature value x).
- We can use the *Bayes formula* to convert the prior probability to the posterior probability $P(w_j|x) = \frac{p(x|w_j)P(w_j)}{p(x)}$

  where $p(x) = \sum_{j=1}^{2} p(x|w_j)P(w_j)$.

Making a Decision

- $p(x|w_j)$ is called the *likelihood* and $p(x)$ is called the *evidence*.
- How can we make a decision after observing the value of x?

$$posterior = \frac{likelihood \times prior}{evidence}.$$

Decide w_1 if $P(w_1|x) > P(w_2|x)$

Decide w_2 otherwise

- Rewriting the rule gives

Decide W1 if $\frac{p(x|w_1)}{p(x|w_2)} > \frac{P(w_2)}{P(w_1)}$

Decide w_2 otherwise

Note that, at every x, $P(w_1|x) + P(w_2|x) = 1$.

Probability of Error

- What is the probability of error for this decision?

$$P(error|x) = \begin{cases} P(w_1|x) & \text{if we decide } w_2 \\ P(w_2|x) & \text{if we decide } w_1 \end{cases}$$

- What is the average probability of error?

$$P(error) = \int_{-\infty}^{\infty} p(error, x)\, dx = \int_{-\infty}^{\infty} P(error|x)\, p(x)\, dx$$

- *Bayes decision rule* minimizes this error because

$$\boldsymbol{P(error|x) = \min\{P(w_1|x), P(w_2|x)\}.}$$

Bayesian Decision Theory

- How can we generalize to
  - more than one feature?
    - replace the scalar x by the feature vector $\mathbf{x}$
  - more than two states of nature?
    - just a difference in notation
  - allowing actions other than just decisions?
    - allow the possibility of rejection
  - different risks in the decision?
    - define how costly each action is

Bayesian Decision Theory

- Let $\{w_1, \ldots, w_c\}$ be the finite set of c states of nature (*classes*, *categories*).
- Let $\{\alpha_1, \ldots, \alpha_a\}$ be the finite set of a possible *actions*.
- Let $\lambda(\alpha_i | w_j)$ be the *loss* incurred for taking action α_i when the state of nature is w_j.
- Let $\mathbf{x}$ be the d-component vector-valued random variable called the *feature vector*.

Bayesian Decision Theory

- $p(\mathbf{x}\,|w_j)$ is the class-conditional probability density function.
- $P(w_j)$ is the prior probability that nature is in state w_j.
- The posterior probability can be computed as

$$P(w_j\,|\mathbf{x}) = \frac{p(\mathbf{x}\,|w_j)P(w_j)}{p(\mathbf{x})}$$

where $p(\mathbf{x}) = \sum_{j=1}^{c} p(\mathbf{x}\,|w_j)P(w_j)$.

Examples

Just to Remember

- $P(w_j \mid x)$ = probability of instance x being in class w_j,

  this is what we are trying to compute

- $P(x \mid w_j)$ = probability of generating instance x given class w_j,

  We can imagine that being in class w_j, causes you to have feature x with some probability

- $P(w_j)$= probability of occurrence of class w_j,

  this is just how frequent the class w_j, is in our database

- P(x)= probability of instance x occurring

  This can actually be ignored, since it is the same for all classes

Sea Bass V. Salmon Classification

- $P(w_j \mid x)$ = probability of instance x being in class w_j,

  this is what we are trying to compute

- $P(x \mid w_j)$ = probability of generating instance x given class w_j,

- **We can imagine that being in class w_j, causes you to have feature x with some probability**

- $P(w_j)$= probability of occurrence of class w_j,

  this is just how frequent the class w_j, is in our database

- $P(x)$= probability of instance x occurring

  This can actually be ignored, since it is the same for all classes

Bayes Classifiers Example

Assume that we have two classes

c_1 = male, and c_2 = female.

We have a person whose sex we do not know, say "*drew*" or *d*.

Classifying *drew* as male or female is equivalent to asking is it more probable that *drew* is male or female, I.e which is greater *p*(male | *drew*) or *p*(female | *drew*)

(Note: "Drew can be a male or female name")



Drew Barrymore



Drew Carey

What is the probability of being called "*drew*" given that you are a male?

What is the probability of being a male?

$$p(\text{male} \mid drew) = \frac{p(drew \mid \text{male})\, p(\text{male})}{p(drew)}$$

What is the probability of being named "*drew*"? (actually irrelevant, since it is that same for all classes)

Bayes Classifiers Example



Officer Drew

This is Officer Drew (who arrested me in 1997). Is Officer Drew a Male or Female?

Luckily, we have a small database with names and sex.

We can use it to apply Bayes rule…

Name	Sex
Drew	Male
Claudia	Female
Drew	Female
Drew	Female
Alberto	Male
Karin	Female
Nina	Female
Sergio	Male

$$p(c_j \mid d) = p(d \mid c_j)\, p(c_j)$$

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)},$$

Bayes Classifiers Example



Officer Drew

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)},$$

Name	Sex
Drew	Male
Claudia	Female
Drew	Female
Drew	Female
Alberto	Male
Karin	Female
Nina	Female
Sergio	Male

$$p(\text{male} \mid drew) = \frac{1/3 * 3/8}{3/8} = \frac{0.125}{3/8}$$

$$p(\text{female} \mid drew) = \frac{2/5 * 5/8}{3/8} = \frac{0.250}{3/8}$$

Officer Drew is more likely to be a Female.

Bayes Classifiers Example



Officer Drew

Officer Drew IS a female!

$$p(\text{male} \mid \textit{drew}) = \frac{1/3 * 3/8}{3/8} = \frac{0.125}{3/8}$$

$$p(\text{female} \mid \textit{drew}) = \frac{2/5 * 5/8}{3/8} = \frac{0.250}{3/8}$$

Bayes Classifiers Example

So far we have only considered Bayes Classification when we have one attribute (the "*antennae length*", or the "*name*"). But we may have many features.
How do we use all the features?

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)},$$

Name	Over 170cm	Eye	Hair length	Sex
Drew	No	Blue	Short	Male
Claudia	Yes	Brown	Long	Female
Drew	No	Blue	Long	Female
Drew	No	Blue	Long	Female
Alberto	Yes	Brown	Short	Male
Karin	No	Blue	Long	Female
Nina	Yes	Brown	Short	Female
Sergio	Yes	Blue	Long	Male

Bayes Classifiers Example

- To simplify the task, **naïve Bayesian classifiers** assume attributes have independent distributions, and thereby estimate

$$p(d|c_j) = p(d_1|c_j) * p(d_2|c_j) * \ldots * p(d_n|c_j)$$

$$p(x \mid w_j) = P(x_1 \mid w_j) * P(x_2 \mid w_j) * \ldots * P(x_n \mid w_j)$$

The probability of class c_j generating instance d, equals….

The probability of class c_j generating the observed value for feature 1, multiplied by.. w_j

The probability of class c_j generating the observed value for feature 2, multiplied by.. w_j

Bayes Classifiers Example

- To simplify the task, **naïve Bayesian classifiers** assume attributes have independent distributions, and thereby estimate

$$p(x \mid w_j) = P(x_1 \mid w_j) * P(x_2 \mid w_j) * \ldots * P(x_n \mid w_j)$$

$$P(\text{officer drew} \mid w_j) = P(\text{over_170}_{cm} = \text{yes} \mid w_j) * P(\text{eye=blue} \mid w_j) * \ldots$$



Officer Drew is blue-eyed, over 170_{cm} tall, and has long hair

$$p(\text{officer drew} \mid \text{Female}) = 2/5 * 3/5 * \ldots.$$
$$p(\text{officer drew} \mid \text{Male}) = 2/3 * 2/3 * \ldots.$$

Bayes Classifier Steps

01	CALCULATE PRIOR PROBABILITY FOR GIVEN CLASS LABELS
02	CALCULATE CONDITIONAL PROBABILITY WITH EACH ATTRIBUTE FOR EACH CLASS
03	MULTIPLY SAME CLASS CONDITIONAL PROBABILITY.
04	MULTIPLY PRIOR PROBABILITY WITH STEP 3 PROBABILITY.
05	SEE WHICH CLASS HAS HIGHER PROBABILITY, HIGHER PROBABILITY CLASS BELONGS TO GIVEN INPUT SET STEP.

Bayes Classifiers Example

- If the features are all independent, then the term $P(X_1, X_2, X_3, \ldots, x_n|C)$ can be re-written as a product of the component probabilities.

According to the following instance decide if the golf will be played or not.
(Outlook= Sunny, Temperature= Hot, Humidity= Normal, wind= Yes).

	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No

Bayes Classifiers Example

- We need to find $P(f_i \mid c_j)$ for each f_i in $\mathcal{F}$ and c_j in $\mathcal{C}$. All these calculations have been demonstrated in the tables below:

Outlook

	Yes	No	P(yes)	P(no)
Sunny	3	2	3/9	2/5
Overcast	4	0	4/9	0/5
Rainy	3	2	3/9	2/5
Total	10	4	100%	100%

Temperature

	Yes	No	P(yes)	P(no)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	9	5	100%	100%

Humidity

	Yes	No	P(yes)	P(no)
High	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	9	5	100%	100%

Wind

	Yes	No	P(yes)	P(no)
False	6	2	6/9	2/5
True	3	3	3/9	3/5
Total	9	5	100%	100%

Play		P(Yes)/P(No)
Yes	9	9/14
No	5	5/14
Total	14	100%

Bayes Classifiers Example

Bayes Classifiers Example

$$P(Yes|today) \propto \tfrac{3}{9}.\tfrac{2}{9}.\tfrac{6}{9}.\tfrac{6}{9}.\tfrac{9}{14} \approx 0.0141$$

and

$$P(No|today) \propto \tfrac{2}{5}.\tfrac{2}{5}.\tfrac{1}{5}.\tfrac{2}{5}.\tfrac{5}{14} \approx 0.0068$$

Now, since

$$P(Yes|today) + P(No|today) = 1$$

- These numbers can be converted into a probability by making the sum equal to 1 (normalization):

43

Bayes Classifiers Example

$$P(Yes|today) = \frac{0.0141}{0.0141+0.0068} = 0.67$$

and

$$P(No|today) = \frac{0.0068}{0.0141+0.0068} = 0.33$$

Since

$$P(Yes|today) > P(No|today)$$

- **So, the prediction that golf would be played is ‘yes’.**

Review Questions

- What is the concept of feature, give an example?
- What is your opinion on this statement, prove your answer "**Adding more features always improves the results of the pattern recognition system**"
- Mention the most used applications of Bayes theory.
- What is the difference between uniform prior and prior probability?
- How can we make a decision with only the prior information?
- Differentiate between prior, class-conditional, and posterior probabilities?.

Review Questions

Assignment_1

According to the example on slide No. 38. If officer drew is blue eyed, over 170 cm tall, and has long hair. Is officer Drew male or female?

Assignment_2

According to the examples in slide No. 43 decide if the golf will be played or not if:-

(Outlook= Overcast, Temperature= Mild, Humidity= Normal, wind= No).

Review Questions

1- Based on Naive Bayes classifier, use the following table to predict if a given instance will be (yes or no in playing tennis) according to the following conditions:

(Outlook =sunny, Temperature= cool, Humidity=high, Wind= strong)

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Review Questions

2- Estimate conditional probabilities of each attribute { colors, legs, height, smiley}for the species class {M, H}using the data given in the table. Using these probabilities to estimate the probability value of the new instance (Color= green, legs= 2, height= tall, and smelly = No).

Day	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

Questions

THANK YOU